Programming Fundamentals II: Lab 05

Due Date: Tuesday October 08, 2024 at Midnight

Part 1. Random Extension (10 Marks)

In Java there exists a Random class that allows us to generate random doubles (as well as some other data types). This class is quite useful, but it may be the case that we want to add functionality to the Random class, without needing to re-write all of complex existing logic that Random implements.

a. **(2 Marks)** Create a new class called RandomUtils that extends the existing Random class. Be sure to accommodate all of the existing constructors for the Random class.
b. **(1 Marks)** Override the nextDouble method of the random class, so that by default it will generate a double between -1 and 1.
c. **(1 Mark)** Create a new version of the nextDouble method that will accept a min and max value, and generate a random double between those values.
d. **(1 Mark)** Create a new method called generateVector that accepts a size and will generate a vector of the given size of doubles between -1 and 1.
e. **(1 Mark)** Overload generateVector so it also accepts a min and max for the doubles it generates.
f. **(1 Mark)** Create a new method called generateMatrix that accepts a length and width that will generate a matrix of the given shape of doubles between -1 and 1.
g. **(1 Mark)** Overload generateMatrix so it also accepts a min and max for the doubles it generates.
h. **(2 Marks)** Create a driver classed called RandomDriver that will use your RandomUtils class to print out
    a. A random integer
    b. A random double
    c. A random double between -100 and 100
    d. A random vector of length 5
    e. A random vector of length 5 with values between -100 and 100
    f. A random matrix of size (2, 4)
    g. A random matrix of size (2, 4) with values between -100 and 100

Add the **date and your name and student number** to the top of your java files. Please submit your java files to Nexus before the end of the day on the due date. **Late submissions will be not be accepted.**